SPHERE
SOFTWARE

TECHNICAL CHALLENGE

AI FULL-STACK DEVELOPER HIRING

Technical Challenges

ROLE	AI Full-Stack Developer	TIME	1–2 hours per challenge
CHALLENGES	1. Address parsing 2. System design	DEADLINE	Rolling

CHALLENGE 1 OF 2: ADDRESS PARSING

Making Sense of Messy Addresses

01 The situation

A home services company has kept customer addresses in its CRM for over a decade. Staff typed them in by hand, in whatever order came to mind, with abbreviations, typos and missing pieces. The company now wants those addresses broken into clean, structured fields.

This challenge tests out-of-the-box thinking. There is no library that will solve it for you, and some of these addresses cannot be parsed at all. How you handle the ones you cannot solve matters as much as the ones you can.

02 The dataset

Thirty addresses in three groups: easy, medium and hard. Each row is a single address string, exactly as staff typed it. Any part of it can be misspelt, abbreviated, in the wrong order or missing. The same data is attached as `addresses.csv`.

Easy 10 ADDRESSES

ID	ADDRESS (AS TYPED IN THE CRM)
A01	Crest Tower.K 74 Opp Harbour View.3, Cuffe Parade, Mumbai 400005
A02	304-Sameer tower/ Hill crest road bandra west, Mumbai 400050
A03	Harbour View 1 B Tower Flat no.57 Cuffe Parade Mumbai 400005.
A04	B-4702 Aurora Kalpa-Park lower parel, Mumbai 400013
A05	Harbour View 1 Flat no.104 A Cuffe Parade, Mumbai 400005
A06	Flat # 214, Marigold Apartment, R.K. Bhosle Road, cuff parade, Mumbai -400005
A07	Crest Tower A 18St Floor flat no Cuffe Parade, Mumbai 400005
A08	Crest Tower.A 11th floor Cuffe parade, Mumbai 400005
A09	Flat No 208 B Nirvana Apt 1 Worli, Mumbai 400030
A10	Nirvana ,IC,/317,Lower Parel, Mumbai 400009

Medium 10 ADDRESSES

ID	ADDRESS (AS TYPED IN THE CRM)
B01	Emerald Heights A1203, JVN Nagar, MHADA, Andheri West, Mumbai 400053
B02	Anjali CHS 503 5th Floor Near Anand Nagar Lane Opp.Metro Mart Chembur Mumbai-71.
B03	B-1603, Oakridge Main St., Hiranandani, Powai, Mumbai 400076
B04	Sarita Building Flat No: 72 7th Floor next to Yb Kamat, Nariman Point, Mumbai 400021
B05	A-706 Orchid Forest Street Hiranandani Powai., Mumbai 400076
B06	B-1302 MAJESTIC HE Nr. Goregaon Bus De Goregaon West, Mumbai 400104
B07	1104 Skyline Sunrise opp Koli Gali Andheri East, Mumbai 400093
B08	Flat no. 902 k wing raheja Horizon Raheja Vihar powai, Mumbai 400072
B09	17C Flat 22 Nalanda Tapovan Caves Road, Andheri East, Mumbai 400093
B10	Sea Havan 3 rd flo Building No. 6 Harbur Road Near Radio club, Mumbai 400005

Hard 10 ADDRESSES

ID	ADDRESS (AS TYPED IN THE CRM)
C01	47, mumbai
C02	Worli, Mumbai 400013
C03	605 D wing Bldg no. 31 MHADA chandivali, Mumbai 400072
C04	C 12 Building No 3 Near Directors Bung IIT Campus, Powai, Mumbai 400076
C05	A/408 Everst Crys Keshavbaug Lane Nr Ashford Colon, Santacruz west, Mumbai
C06	Harbour View.1 A Flat No.46, Cuffe Parade, Mumbai 400005
C07	T2-1804 K & R Sapphire isle Powai Brightpath Technologies Pvt Ltd. C-2215-18 Silverline Garden Estate, Mumbai
C08	E wing 1507 vasnt splendr jvlr, Jogeshwar West, Mumbai
C09	07 3rd Floor of a Building Nova Kiran CHSL Road Landmark: Frosty Cones Ice Cream Lane, Mahim, Mumbai 400016

03 What your code should do

Write a script that reads the dataset and, for every address, outputs five fields: **building name, street, area, city and pin code**. If a field does not exist in the address, leave it blank. Do not invent values.

The hard group cannot be parsed reliably. Your code must detect addresses like these and flag them instead of guessing. For every flagged address, output which fields are missing or unreliable and why. For example: “no building name”, “two addresses in one field”, or “pincode missing”. Your code should catch these on its own, not because it knows which ID is in which group.

Questions to answer in your README

- What should the correct output be for each address? Where you are unsure, say what you would output and why.
- Are there any other fields you think the output should include? Justify each one.
- What rule decides whether an address is flagged rather than parsed?
- What would change if this ran across 20,000 addresses instead of 30?

04 What to submit

- Your code, with instructions to run it.
- The output as a CSV: the five fields, plus a flag column and a column explaining what is missing.
- A short README answering the questions above.

05 How this is assessed

- Whether your code knows when *not* to guess.
- Whether your flags say clearly what is missing, not just that something is wrong.
- Whether your approach would still work on addresses you have not seen.
- The reasoning in your README.

CHALLENGE 2 OF 2: SYSTEM DESIGN

Automating a 12-Year-Old CRM

Written design. No code required.

01 The client

Shine Home Services is a home cleaning company operating across Mumbai. Customers book one of three packages, each quoted individually by the company based on the size of the home, the number of rooms, and other factors. There is no fixed price list.

PACKAGE	WHAT IT INCLUDES
1. Simple	Floors and windows only.
2. Medium	Floors and windows, beds made, all surfaces wiped down.
3. Deep	Everything in Medium, plus cleaning underneath tables and moving furniture.

Most customers book on a recurring basis: weekly, fortnightly or monthly. The company grows revenue primarily by upselling add-on services to these existing customers: closet organisation, mattress deep clean, upholstery cleaning, and similar one-off jobs.

02 What you are working with

The client runs a custom CRM and ERP system built by a different vendor roughly twelve years ago. It handles simple entries and little else. Assume no API, no webhooks, no documentation, and no support from the original vendor.

Access you have been given

- No access to the client's production server.
- A server holding a **mirror of the production MySQL database**, with a MySQL username and password, plus root access to that mirror server.
- A login to the CRM/ERP web application itself, with browser access only.
- The option to choose and provision your own server for development work.

Manual exports available from the CRM interface

- **Sales report**: filtered by a date range you select.
- **Bills report**: invoices that have been created.
- **Customers report**: customer details only.

> A system of this age is rarely clean. Timestamps may be missing, edits may overwrite history, statuses may be free text, and the same customer may exist several times over. Anything you do not know, write down as an assumption and design around it.

03 The five automations

The client wants five automations built on top of this database. Describe how you would deliver each one.

01

Billing report to the accounts team

Billing data from the CRM, emailed to the accounts team automatically. Decide what the email contains, how often it goes out, and what happens when the data looks wrong.

02

Change log

A list of what changed in the database: customer details edited, bills manually modified, records added or removed. Consider how you detect a change in a system that may not record one.

03

Outstanding bills

Bills that have been generated but not yet paid by the customer. Consider how ageing is presented and who needs to see it.

04

Bank statement versus CRM reconciliation

Match payments appearing in the bank statement against bills in the CRM. Explain your matching approach **and the edge cases it has to survive**. This is the automation we will read most closely.

05

Founder report (open ended)

The company wants to grow revenue and has asked you for insights. Decide what a founder of this business should see, and why those particular numbers would change a decision. Keep the recurring-booking and add-on upsell model in mind.

04 What to submit

SECTION	WHAT WE EXPECT
Architecture	The infrastructure you would put in place. A diagram or a clear written description: where data comes from, what processes it, where it is stored, how each automation is triggered, and what happens when something fails.
Server & cost	The specs of the server you would use, with a real provider, a real price and a link to the pricing page. Say why that size and not a larger or smaller one.
Assumptions	Everything you had to assume because the brief does not say. List these explicitly.
Approach per automation	A short section on each of the five, including edge cases for the reconciliation.
Three questions	The three questions you would ask the client before writing any code. Choose the ones whose answers would change your design the most.

05 How this is assessed

- Whether your design is realistic about what a twelve-year-old system can and cannot do.
- Whether your assumptions are stated rather than hidden.
- The quality of your thinking on reconciliation edge cases.
- Restraint: the simplest infrastructure that does the job beats the most impressive one.
- Whether your three questions are the ones that would genuinely unblock the work.

There is no single correct answer. We are reading for judgement, not for a particular stack.

06 Submitting both challenges

Reply to the email you received this brief in, with a single **ZIP file** attached. The ZIP should contain:

- **Challenge 1:** your code, the output CSV and your README
- **Challenge 2:** your write-up as a PDF or Word document
- **Optional:** your full AI chat logs, as exported files or screenshots

On using AI

Using AI tools is expected and encouraged. It is how we work. If you attach your full chat logs alongside your submission, that counts in your favour. We are interested in how you direct the tool, where you disagreed with it, and what you checked before accepting an answer.

Submissions are reviewed on a rolling basis. The earlier you submit, the sooner you can move to the next stage, which is an interview.

Each challenge should take one to two hours. If you run out of time, submit what you have with a note on what you would do next.